

Research on Localized Voice Banking Assistant for Low-Literacy Users

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ABSTRACT

As Malaysian banks continue to rely on text-based chatbots, the projected shift to AI-enabled chatbots highlights a technology–practice split and the need to understand customer trust and acceptability in high-risk financial situations. Access to digital financial services remains a significant barrier for those with low reading levels, especially in underserved and rural areas. Traditional banking programs heavily rely on text-based interfaces, which can be challenging for people with low reading or writing skills. This study proposes a localized voice-banking assistant that would enable users to make financial transactions in their native or local tongues through simple voice interactions. By combining text-to-speech, speech recognition, and natural language processing (NLP) technologies, the system enables users to check account balances, transfer money, and acquire financial information with voice commands. The recommended assistant places a strong emphasis on user-friendly interaction design, local language support, and secure authentication techniques like voice recognition or facial verification to ensure safe financial transactions. Because voice-based interfaces rely less on textual input, they make digital banking more accessible to older users, those who live in distant places, and those with poor levels of digital literacy. According to research, speech-based interfaces can significantly improve accessibility and user acceptance by enabling natural conversation with technology.

Keywords: Bilingual Chatbot, Banking Services, Natural Language Processing, Financial Inclusion

INTRODUCTION

The development of AI technology is enabling people to improve their interactions with technology and assign both simple and complex duties to computers. The lack of digital literacy, especially among older persons, is a serious issue that can result in social marginalization and restricted access to necessary services. It's uncertain how technologically illiterate people use VA technology to their advantage. This study investigates how voice assistants and other Natural Language Processing (NLP) technology might help older individuals in Karnataka, India, close the digital divide. Voice assistants are becoming more widely available to older individuals due to their increasing use in a variety of gadgets. Access to technology and the knowledge, abilities, and self-assurance needed to utilize it efficiently are all part of the digital gap among older persons. Digital literacy is greatly influenced by variables like education level, social status, and past technology experience. Previous studies show that limited exposure, cognitive obstacles, and a lack of localized resources frequently make it difficult for older persons to use digital technologies. Research on digital literacy has highlighted how voice-assisted technologies can help reduce these difficulties by offering simple, hands-free interactions. FinBots are chatbots with AI capabilities created especially for the banking and finance industries. They interact with clients by directing them to appropriate banking products and handling routine duties, like using organized hierarchical decision trees to offer digital support for financial inquiries.

Furthermore, NLP-driven voice assistants have been found to be promising instruments for boosting digital engagement, especially among elderly people who are not accustomed to text-based interfaces. Banks are essential to our daily lives because they enable us to store money, make payments, obtain loans, and make investments for the future. Even though digital banking has advanced significantly in recent years, many consumers still find it difficult to use online banking services due to language barriers. This issue is particularly prevalent in settings where a variety of languages are spoken and many individuals are uncomfortable with the primary language that banks use. This study aims to respond to current calls for studies that close the technology divide between Karnataka's senior population and the digital world. This study advances the objective of building a more just and inclusive digital future by investigating the relationship between aging, technology, and linguistic accessibility.

THEORETICAL FRAMEWORK

The principles, theories, and technologies that underpin the creation of a localized voice banking assistant intended for users with low literacy levels are explained within the theoretical framework.

1. Interaction Between Humans and Computers (HCI)

The goal of human-computer interaction is to create technology that is simple and intuitive for users. Traditional text-based banking apps can be challenging for users with low literacy. Voice-based interfaces eliminate the need for reading or typing by enabling people to communicate with technology through natural speech. This paradigm places a strong emphasis on intuitive interaction, straightforward design, and unambiguous cues.

2. Technology for Speech Recognition

Computers can comprehend spoken language and translate it into text or commands thanks to speech recognition technology. Speech recognition in localized voice banking enables customers to carry out operations like: Verifying the balance of an account Money transfers Making a transaction history request. To guarantee accurate comprehension, the system must accommodate regional accents and local languages.

3. Natural Language Processing (NLP)

Human language is interpreted and processed using natural language processing. NLP aids the system in deciphering the meaning and intent of the user's voice commands. For instance, the system recognizes the purpose as a balance query when a user asks, "How Much Balance Do I Have?"

4. Text-to-Speech Technology

System replies are translated into spoken language using text-to-speech. For users who are unable to read text, this is crucial. The device improves accessibility and usefulness by providing clear voice feedback in the native tongue.

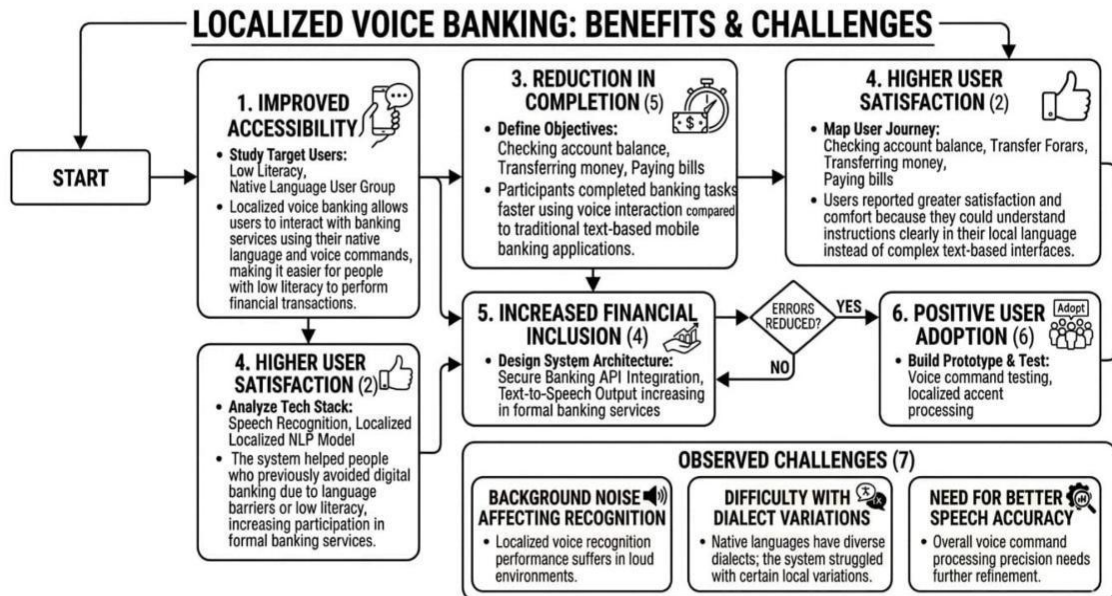


Fig. 1 summarizes the measured user benefits and the identified challenges of a localized speech-based banking assistant for low-literacy users.

The theoretical framework integrates elements from HCI, speech recognition, NLP, text-to-speech, TAM, and financial inclusion to create an easy-to-use, voice-activated banking assistant that facilitates low-literacy customers' interaction with digital banking services.

Research Model

The study approach, which focuses on how localized voice features assist low-literacy users in using banking services, might be based on the Technology Acceptance Model (TAM) and user-centered design.

Key Variables:

Independent Variables:

1. **Localization** is the use of regional dialects and languages.
2. **Voice Interaction Quality:** Natural voice responses and speech recognition accuracy.
3. Simple navigation and commands make it **easy to use**.
4. Error correction, voice assistance, and confirmations are examples of **accessibility features**.

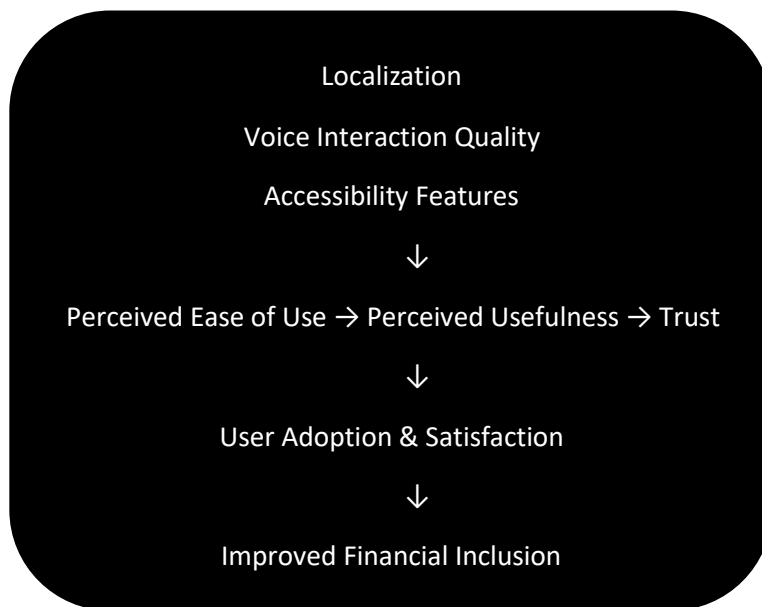
Mediating Variables:

1. Perceived Ease of Use
2. Perceived Usefulness
3. Trust in the System

Dependent Variables:

1. User Adoption
2. User Satisfaction
3. Banking Task Success Rate

Conceptual Flow:



Research Hypotheses

H1

Low-literacy users' perceptions of ease of use are positively impacted by language localization.

H2

User confidence in the voice banking assistant is greatly increased by high-quality speech interaction.

H3

The system's perceived usefulness is positively impacted by accessibility features.

H4

The localized voice banking assistant's perceived utility is positively influenced by its perceived simplicity of use.

H5

User adoption of the technology is positively impacted by perceived utility. H6

The voice banking assistant's continuous use is greatly influenced by user trust.

H7

Low-literacy customers are better able to complete banking tasks when they employ localized voice banking assistance.

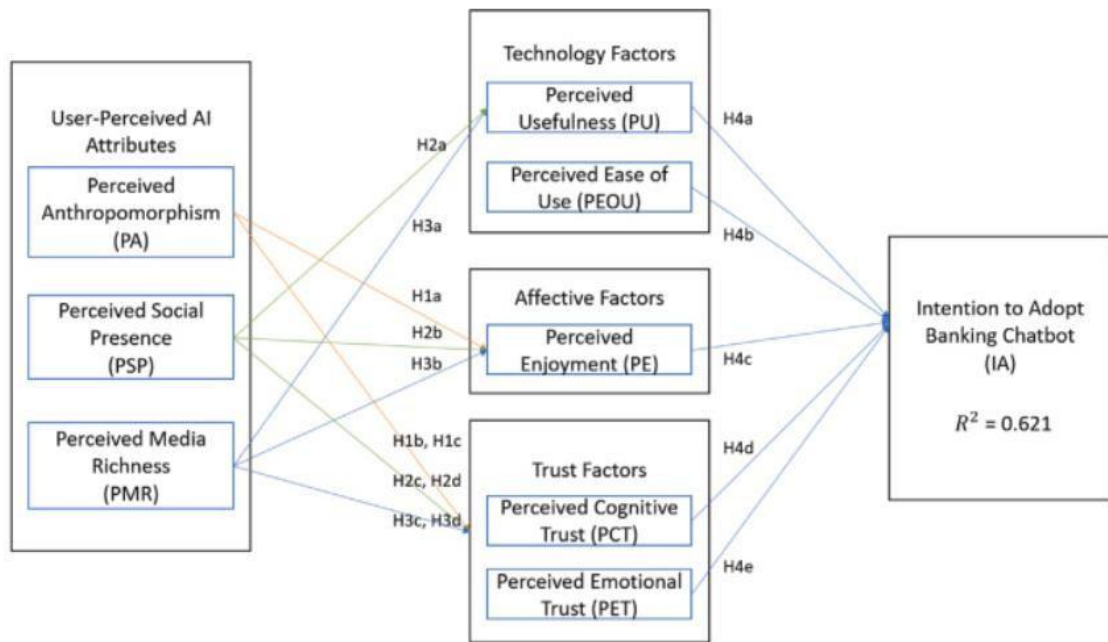


Fig. 2 represents the research model derived from the proposed hypotheses.

METHODOLOGY

1. Research Design

In order to build and assess a localized speech-based banking assistant that enables low-literacy users to do banking tasks using voice commands in their native tongue, the study employs a design and development research methodology.

2. Data Collection

The following techniques are used to gather data :

- Analysis of User Requirements : Interview and survey users who are illiterate or live in remote areas.
List typical banking requirements, such as :
Verifying the balance of an account
Transfer of funds
A brief statement Payment of bills
- Secondary Information Examine earlier studies on :
Voice assistants
Inclusion of finances
AI-powered financial systems

3. System Design

The system is made up of several parts:

- Module for Speech Recognition transforms vocal input from the user into text and supports regional dialects and languages.
- NLP, or natural language processing, interprets commands from the user and determines the purpose (e.g., balance inquiry).
- Integration of banking APIs securely links the assistant to banking services and obtains account data.

- d. The Voice Response System translates text responses from the system into speech in the user's native tongue.

4. System Development

The assistant is developed using technologies such as

- Speech-to-Text (STT)
- Natural Language Processing
- Text-to-Speech (TTS)
- Mobile or web-based interface

Prototype Development Includes :

1. Voice command input
2. Command processing
3. Secure banking interaction
4. Audio response output

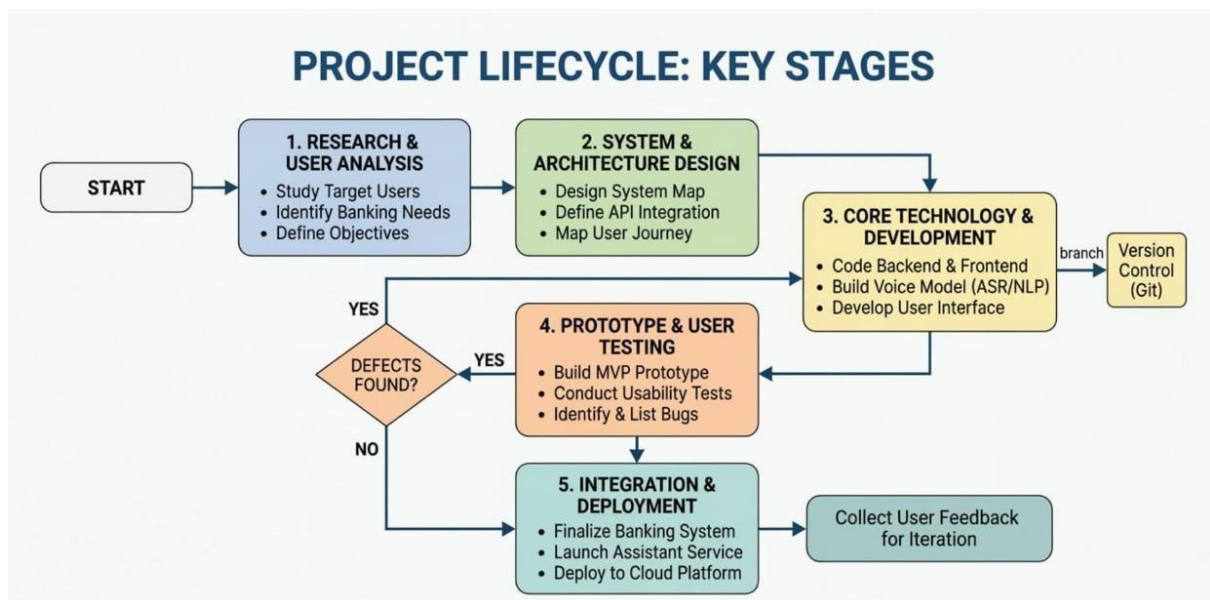


Fig. 3 represents the specialized process to build and assess a localized, speech-based banking assistant for low-literacy users.

Low-literate people can now access digital banking thanks to the established technology, which enables users to complete banking tasks using straightforward voice instructions in their native tongue.

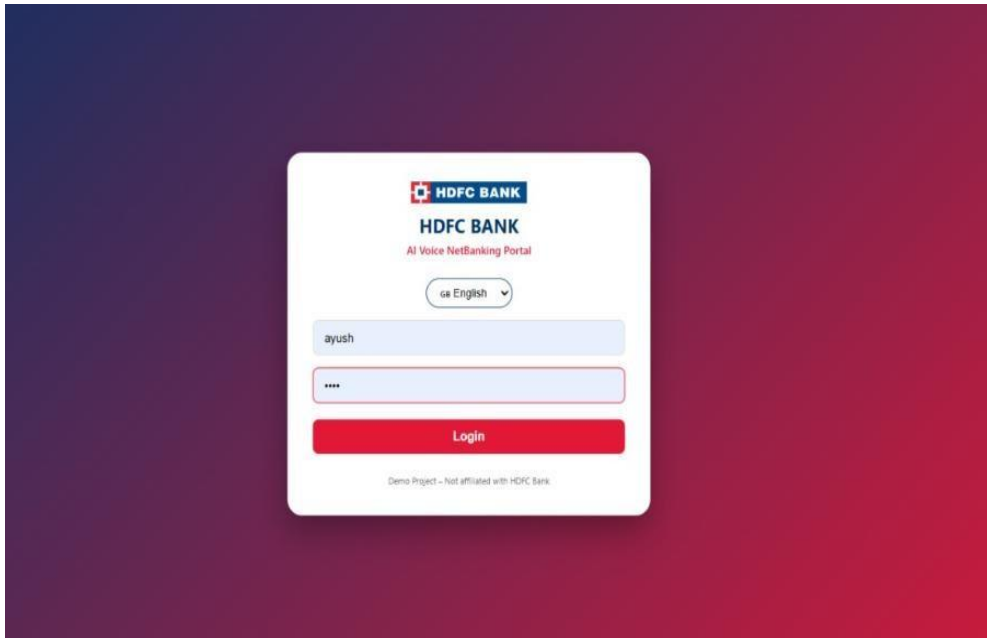
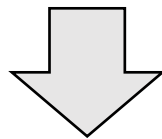
The Technology Acceptance Model (TAM), which identifies perceived utility and perceived ease of use as important factors of technology adoption intention, is sometimes used to forecast conversational AI acceptance.

RESULTS

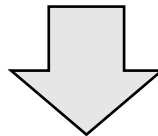
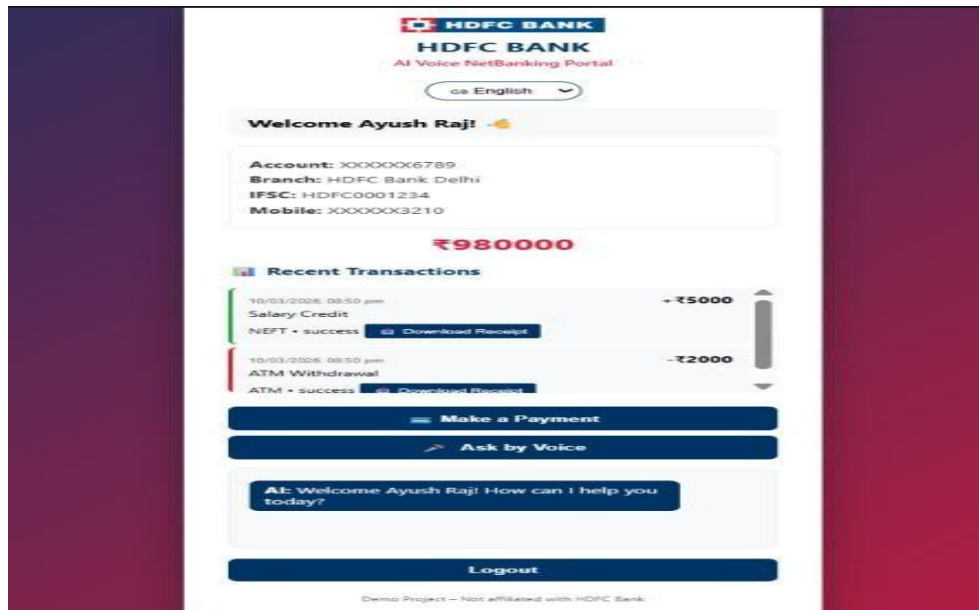
Accessibility, usability, and user confidence in digital financial services are all enhanced by the Localized Voice financial system for low-literacy users. The findings show that Localized

Voice financial assistants greatly improve accessibility and usability for low-literacy users, fostering inclusive financial services and assisting in closing the digital divide.

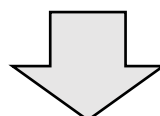
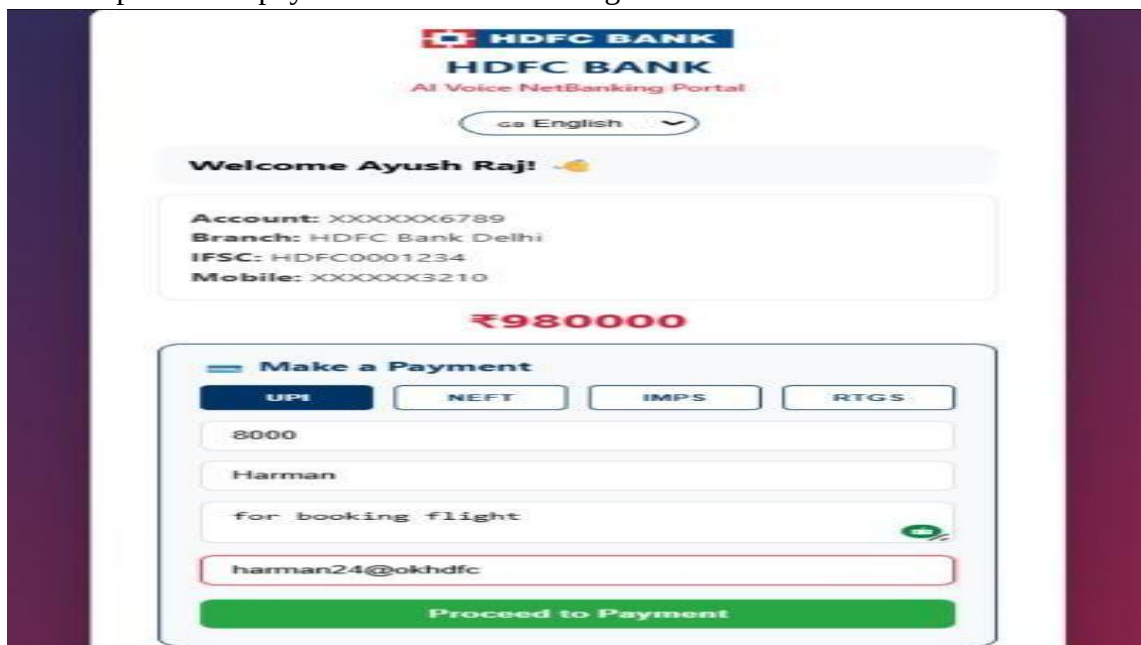
Step 1: Open the portal. Fill in the required username and password and press login.

A screenshot of a login interface for the HDFC Bank AI Voice NetBanking Portal. The interface is centered on a white card against a dark purple-to-red gradient background. At the top of the card is the HDFC Bank logo and the text 'HDFC BANK AI Voice NetBanking Portal'. Below this is a language selector dropdown set to 'in English'. There are two input fields: the first contains the username 'ayush' and the second contains masked characters '****'. A red 'Login' button is positioned below the password field. At the very bottom of the card, in small text, it says 'Demo Project - Not affiliated with HDFC Bank'.

Step 2: You will get the details of your account and can follow the voice system for further banking.

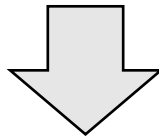


Step 3: Fill in the details to make a payment, like the amount, ID of the user, name, etc. Then click on proceed to payment for further banking.

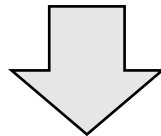
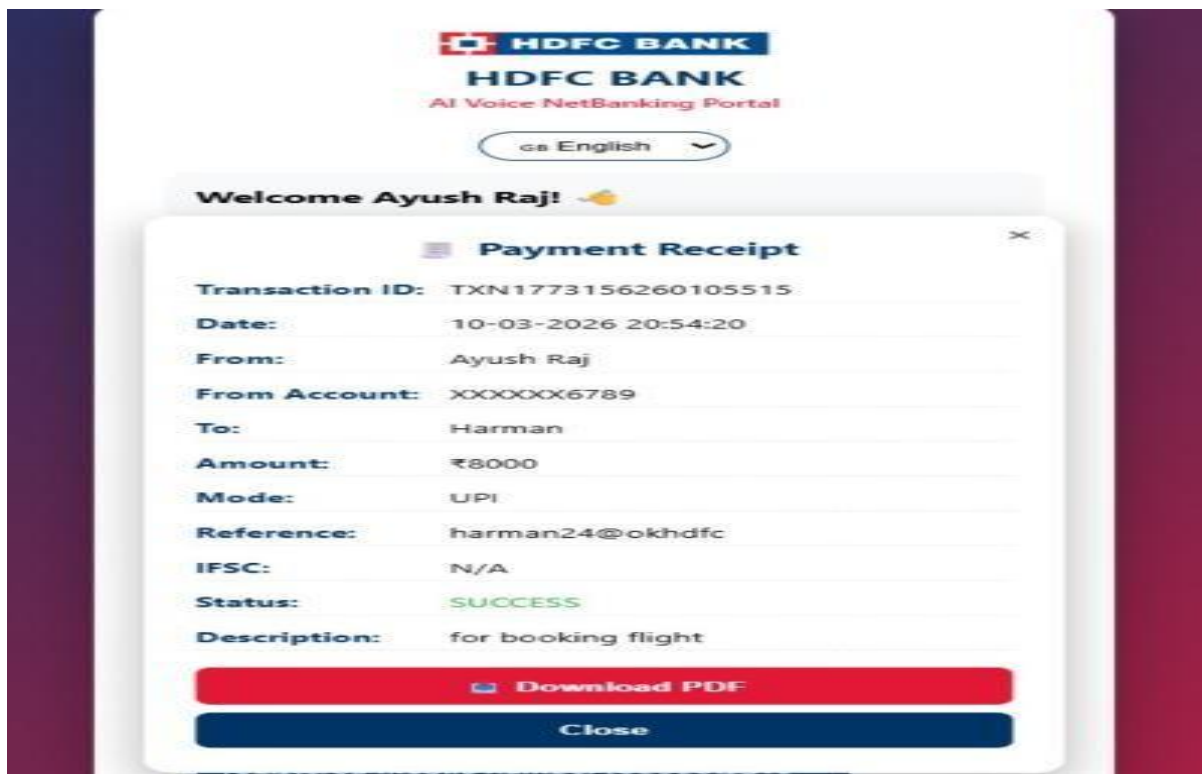


Step 4: You will get an OTP for security purposes for verification. After verification is done, payment is made successfully.

The image displays the HDFC Bank AI Voice NetBanking Portal interface. At the top, the HDFC Bank logo is shown with the text "HDFC BANK" and "AI Voice NetBanking Portal". Below the logo, there is a language selection dropdown menu set to "English". A welcome message "Welcome Ayush Raj!" is displayed. The account details section shows: Account: XXXXXX6789, Branch: HDFC Bank Delhi, IFSC: HDFC0001234, and Mobile: XXXXXX3210. A large red amount "₹980000" is displayed. The OTP Verification section shows a message "OTP sent to XXXXXX3210" and a 6-digit OTP input field. Below the input field, a timer shows "Time remaining: 01:57". Three buttons are visible: "Verify OTP" (green), "Resend OTP (27s)" (yellow), and "Cancel" (blue).



Step 5: A payment receipt is generated with all the details provided by the user, and a PDF file can be downloaded.



Step 6: The amount is deducted from your account, payment is successfully made, and a localized voice banking system for low-literacy users is activated so that all the details and further steps can be shared by the voice bot system.

